

Turf Pad Build Spec — 13 ft x 24 ft Cut-and-Fill Terrace

Leveling ~6 ft of grade across a 13 ft depth: cut 3 ft at the uphill (back) edge, fill 3 ft at the downhill (front) edge. Pad surface = 312 sq ft. Each wall runs the full 24 ft width.

Earthwork (cut and fill roughly balance)

Item	Quantity	Notes
Cut (back wedge)	~8.7 cu yd	Excavate down 3 ft at back, tapering to 0 at center
Fill (front wedge)	~8.7 cu yd	Build up 3 ft at front, tapering to 0 at center
Net import	~0 (balanced)	Native cut can become fill if soil is clean & non-organic

- Excavated soil swells ~20–25%, then compacts back. Place fill in **6–8 in lifts**, compact each lift to ~90–95% with a plate or jumping-jack compactor.
 - If native soil is clayey/organic/soft, import engineered fill or crushed base for the load-bearing portion instead of reusing it. The front (fill) side is structural — don't cut corners here.
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Walls — sized separately (they do different jobs)

Both: segmental retaining block (SRW), 3 ft exposed, ~6 in embedded base course (~10% of height), 24 ft long.

FRONT wall — holds engineered FILL (the critical one)

Item	Qty	Notes
SRW blocks (~1 sq ft face each)	~84	~24 ft x ~3.5 ft face

Cap blocks	~16	adhesive-set
Geogrid reinforcement	~250 sq ft	2 layers, ~5 ft deep, tied back into compacted fill
Leveling pad (crushed base)	~0.9 cu yd	6 in × 24 in × 24 ft, compacted
Drainage rock (¾" clean)	~2.7 cu yd	12 in chimney behind wall, full height
4" perforated pipe	~24 ft + outlet	base of drainage chimney
Filter/separation fabric	~190 sq ft	wrap the drainage rock

Surcharge check: this wall sits above the existing pool wall. Keep it **structurally independent** — its footing/load must not bear on the pool wall.

BACK wall — retains the CUT face (undisturbed native soil)

Item	Qty	Notes
SRW blocks	~84	same geometry
Cap blocks	~16	
Geogrid	usually none/minimal	gravity wall typically OK at 3 ft into native cut — verify with block manufacturer's chart
Leveling pad (crushed base)	~0.9 cu yd	
Drainage rock (¾" clean)	~2.7 cu yd	still required — intercept hillside water
4" perforated pipe	~24 ft + outlet	this is your interceptor drain
Filter fabric	~190 sq ft	

Turf base (over the leveled subgrade, 312 sq ft)

Layer	Material	Qty

Separation / weed barrier	woven geotextile	~360 sq ft
Sub-base, 4 in compacted	Class II road base ($\frac{3}{4}$ "-minus)	~5 cu yd (incl. compaction allowance)
Bedding, 1 in	decomposed granite or sharp sand	~1 cu yd

Build the base with ~1–1.5% fall toward the front collector drain. Compact in lifts.

Artificial turf + finishing

Item	Qty	Notes
Turf	15 ft wide roll × 24 ft = ~360 sq ft	15 ft width covers the 13 ft depth seam-free; ~13% waste
Infill (silica/coated sand)	~500 lb (10 × 50 lb)	~1.5 lb/sq ft, brushed in
Galvanized turf spikes (5–6")	~1 box (~250)	perimeter (~74 lin ft) + any seams
Seam tape + adhesive	as needed	only if you seam

Aggregate shopping summary (round up)

- Class II road base: **~7 cu yd** (5 pad + ~2 for two wall leveling pads)
- $\frac{3}{4}$ " clean drainage rock: **~5.5 cu yd** (both wall chimneys)
- DG / sharp sand bedding: **~1 cu yd**

Tools (rent)

Plate compactor or jumping-jack · transit/laser level or string lines · wheelbarrow · turf knife/cutter · power broom for infill.

Before you dig

- **Call 811 / DigAlert** to locate utilities.
- Walls under 4 ft (footing-to-top) usually need **no permit** in most CA jurisdictions — BUT a *fill-retaining* wall, hillside lots, and proximity to a pool can trigger engineering or permits. The two walls are ~13 ft apart ($> 2\times$ their height), so they're likely treated independently, not as a tiered system. **Confirm with your local building department.**